

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n,x,floor = -1;
4      scanf("%d",&n);
5      int a[n];
6      for(int i=0;i<n;i++)
7          scanf("%d",&a[i]);
8
9      scanf("%d",&x);
10     for(int i=0;i<n;i++){
11         if(a[i]<=x)
12             floor=a[i];
13         else
14             break;
15     }
16     printf("%d",floor);
17     return 0;
18
19 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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